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Juhyun Park

Curriculum Vitae

Instructional Systems Technology, Indiana University Bloomington
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EDUCATION

Ph.D., Instructional Systems Technology Aug 2024 – Present
Indiana University Bloomington
Minor: Qualitative and Quantitative Research Methodology
First Advisor: Dr. Kyungbin Kwon, Second ADvisor: Dr. Ugur Kale

M.Ed., Elementary Mathematics Education Mar 2021 – Aug 2023
Seoul National University of Education, South Korea
Thesis: *The Effects of Mathematics Class Using Virtual Reality on Elementary School Students' Spatial Ability and Mathematical Attitudes*

B.Ed., Elementary Education Mar 2015 – Feb 2019
Seoul National University of Education, South Korea

RESEARCH PROJECTS

Independent Research

AI-Supported Lesson Planning for Pre-Service Teachers Fall 2025 – Present
Design-based research, Indiana University Bloomington | Role: Lead Researcher (first-author study)

- Identified pre-service teachers' lesson-planning challenges through a literature review and interviews with experienced instructors (3+ years) and undergraduate learning assistants.
- Designed and prototyped an AI tool that scaffolds pre-service teachers' self-regulation during lesson planning, aiming to prevent cognitive offloading of novices with limited pedagogical content knowledge (PCK).
- Conducting iterative design cycles: Prototype 1 implemented in two course sections (Spring 2026) and a refined Prototype 2 in three sections (Fall 2026); findings to be presented at AECT 2026.

NSF-Funded Projects

NSF SILICON Project Fall 2025 – Present
Collaborative Research: Semiconductor Informal Learning Initiative through Community-Oriented Networks
PI: Kyungbin Kwon | Role: Graduate Research Assistant

- Designing a sequence of hands-on, embodied lessons that make the computing foundations of semiconductors tangible for middle school students: binary numbers, logic gates, and binary addition with logic gates.
- Structuring the lessons as community-based sessions in which students learn together with their families, extending the curriculum to informal learning settings.
- Presented the lesson design at the CRLT Graduate Student Poster Session (2026); curriculum currently in development.

NSF AI Goes Rural Project Fall 2024 – Spring 2026
PI: Kyungbin Kwon | Role: Graduate Research Assistant

Ben Davis University High School, Indianapolis, IN — High School CS *Fall 2025 – Spring 2026*

- Co-designed and studied a project-based learning unit with a CS teacher in which students explored how AI could solve a local school bus route problem.

Bloomfield Elementary School, Bloomfield, IN — Grade 2 *Spring 2025*

- Developed instructed vs spontaneous embodied computational thinking activities; collected and analyzed data and co-authored the manuscript.

Jasper High School, Jasper, IN — High School ELA *Fall 2024 – Spring 2025*

- Designed four generative AI chatbots for units on argumentative writing (claim, evidence, and warrant), persuasive writing, and career preparation (résumé writing and mock interviews).
- Collected and analyzed data from 47–57 students per unit, including student–AI interaction logs, written reflections, AI literacy post-surveys.
- Examined students' AI literacy, cognitive engagement, and attitudes toward AI feedback; co-authored journal articles, a proceedings paper, and a manuscript in preparation.

Collaborative Research Projects

Robot Arm Project, with National Taiwan Normal University Spring 2026
Binford Elementary School, Bloomington, IN — Grade 5 Science

- Designed a two-lesson sequence connecting computational thinking and AI literacy to environmental issues: Lesson 1 used Teachable Machine to explore how machine learning works, and Lesson 2 combined MediaPipe and a robot arm for embodied AI learning.
- Co-implemented the lessons with three Grade 5 science teachers at Binford Elementary.

CSforGood Project Fall 2025; Fall 2026
Childs, Lakeview, and Unionville Elementary Schools, Bloomington, IN

- Facilitated project-based learning in which elementary students applied CS and AI to solve problems in their communities, across three schools over two implementation cycles (Nov–Dec 2025; Oct–Nov 2026).
- Collected student artifacts and classroom observations, and conducted interviews with three in-service teachers.

Grant Writer / Project Lead (South Korea)

STEAM Teachers' Study Group 2022 – 2023
Korea Foundation for the Advancement of Science and Creativity (KOFAC), $4,000

- Wrote the proposal and led teachers in designing, implementing, and collecting data on STEAM programs for 549 students.

Online Teaching and Learning Materials for ITDA 2022
Korea Education and Research Information Service (KERIS), $6,000

- Led teachers from six schools in developing nationwide online content based on student and teacher interviews; delivered teacher PD.

Innovative Teaching Methods Research 2021, 2022
Seoul National University of Education, $1,000 for each

- Developed blended and AI-enhanced learning materials informed by a survey of 445 students and 19 teachers.

PUBLICATIONS

Peer-Reviewed Journal Articles

1. Kwon, K., Yang, S., Kale, U., Kim, K., & **Park, J.** (2026). Generative AI in high school English career preparation units: Student interactions, perceptions, and ethical concerns. *Computers and Education: Artificial Intelligence, 10*, 100588. https://doi.org/10.1016/j.caeai.2026.100588

2. Kwon, K., Kale., U., Kim, K.& **Park, J.** (in preparation). Embodied learning for computational thinking in early elementary students: Effects of instructed vs. spontaneous approaches. (Under Review)

3. Kwon, K., **Park, J.**, Yang, S., Kale, U., & Kim, K. (in preparation). Generative AI for writing: Student interactions with AI feedback in a high school English classroom. (Under Review)

Book Chapters

1. Ugur, K., Kwon, K., Ozogul, G., & **Park, J.** (2025). An inquiry-based computational thinking approach in teacher education. In *Advancing Teacher Education via Innovative Pedagogies* (pp. 141–180). IGI Global. https://doi.org/10.4018/979-8-3373-2660-3.ch005

2. **Park, J.** (2023). How to design curriculum with generative AI. In *We Walk with Schools: Opening Education Communities and Future Learning* (pp. 99–101). Seoul Seobu District Office of Education. (in Korean)

3. **Park, J.** (2022). A project to create eco-friendly buildings in a virtual reality world. In *Yes, You Can! Blended Learning Materials Collections* (2nd ed., pp. 194–198). Seoul Metropolitan Office of Education. (in Korean)

Peer-Reviewed Conference Proceedings

1. Kwon, K., Kim, H., Kim, K., Yang, S., & **Park, J.** (2025). AI education in English class: Exploring benefits, challenges, and ethical implications of using generative AI for argumentation. In *Proceedings of EdMedia + Innovate Learning* AACE.

2. **Park, J.** (2023). The effects of mathematics class using virtual reality on elementary school students' spatial ability and mathematical attitudes. In *Proceedings of the 60th Conference of the Korea Society of Educational Studies in Mathematics* (pp. 149–150).

PRESENTATIONS

Peer-Reviewed Conference Presentations

1. **Park, J.**, & Kwon, K. (2026, November). *From efficiency to agency: Design-based research on dialogic AI-supported lesson design.* Association for Educational Communications and Technology (AECT) International Convention, Chicago, IL.

2. **Park, J.**, Kwon, K., Yang, S., Kim, K., & Kale, U. (2026, November). *Exploring AI-supported writing in high school English classrooms: Student engagement, judgment, and perceptions of AI feedback.* Association for Educational Communications and Technology (AECT) International Convention, Chicago, IL.

3. Yang, S., **Park, J.**, Park, D., Kale, U., & Kwon, K. (2026, November). *Exploring computer science project-based learning: Enacted PBL elements and teacher and student growth.* Association for Educational Communications and Technology (AECT) International Convention, Chicago, IL.

4. Yang, S., Kwon, K., Kale, U., Kim, K., & **Park, J.** (2026, November). *Learning with generative AI in high school English classes: Student interactions, perceptions, and ethical concerns in career preparation lessons.* Association for Educational Communications and Technology (AECT) International Convention, Chicago, IL.

5. Hong, S., **Park, J.**, Kwon, K., & Lim, C. (2025, October). *How artificial intelligence is integrated in education: Objectives, strategies, and outcomes—A systematic literature review* [Poster]. Association for Educational Communications and Technology (AECT) International Convention, Las Vegas, NV.

6. **Park, J.** (2025, April). *Design principles to mitigate over-reliance in pre-service teachers' lesson design with a generative AI chatbot.* Society for Information Technology and Teacher Education (SITE), Philadelphia, PA.

7. **Park, J.** (2023, June). *The effects of mathematics class using virtual reality on elementary school students' spatial ability and mathematical attitudes.* Korea Society of Educational Studies in Mathematics, Seoul, South Korea.

Conference Presentations

8. **Park, J.** (2026, September). *Be the bit, be the gate: How computers count and calculate* [Poster]. Center for Research on Learning and Technology (CRLT) Graduate Student Poster Session, Bloomington, IN.

9. **Park, J.** (2025, March). *A framework for integrating computational thinking into pattern generalization in elementary mathematics education.* Instructional Systems Technology (IST) Conference, Bloomington, IN.

10. **Park, J.** (2022, October). *Developing a web-based VR instructional program for elementary mathematics* [Poster]. SNUE Academic Festival, Seoul, South Korea.

11. **Park, J.** (2022, January). *Developing Blended learning materials for overpopulated classes- integrating Korean and Mathematics Classes.* Presented at the Seoul National University of Education, Seoul, Korea.

Invited Talks

1. **Park, J.** (2023, July). *How to effectively make use of generative AI in my class*, Presented at the Seoul Seobu District Office of Education, Seoul, Korea.

2. **Park, J.** (2023, June; 2022, June; 2018, November; 2017, November) *How to become a successful teacher specialized in mathematics and technology in elementary school.* Presented at the Jinmyeong Girls High School, Seoul, Korea.

3. **Park, J.** (2022, December). *Achievement of developing an online teaching and learning materials for ITDA.* Presented at the Korea Institute of Science and Technology Information, Jeju, Korea.

TEACHING

Higher Education Undergraduate Course Teaching

Associate Instructor, Indiana University Bloomington

2025 Fall, 2026 Spring (3session) EDUC-W200: Teaching with Technology

A project-based course for pre-service teachers on technology integration, covering inclusion and diversity, technology for productivity, and makerspaces. Students design technology-integrated lesson plans and build professional ePortfolios.

Teacher Professional Development

Oct, 2026 AI, Machine Learning, and Teachable Machine for Grade 5 Science Teachers, Binford Elementary School, Bloomington, IN

Feb, 2025 AI in Indiana Teacher Professional Development, Indianapolis, IN

2023–2024 District-wide generative AI and ed-tech workshops, Leading Teacher of Educational Technology, Seoul Seobu District Office of Education

K–12 Classroom Teaching (2019–2024), Seoul, South Korea

Bakseok and Gajaeul Elementary School Teacher, Seoul, Korea Mar, 2019 – Aug, 2024

- Taught Grades 4 and 6 across core subject areas in public elementary schools.
- Designed and implemented interdisciplinary and locally responsive curricula as part of an innovation-school initiative.
- Designed and led AI and computing programs focused on computational thinking, AI ethics, robotics, and programming.
- Developed differentiated after-school programs to support students requiring additional academic support.
- Led hands-on STEM and technology learning activities that connected computing concepts with students' everyday learning experiences.

STEM-Gifted Education Teacher, Seoul Seobu District, Korea Mar, 2021 – Dec, 2023

- Designed and taught mathematics, science, and computing enrichment courses for gifted students in Grades 4–5.
- Developed inquiry- and problem-based learning activities in computational thinking, spatial reasoning with tangrams, centroids, and introductory machine learning.
- Integrated physical manipulatives, computational tools, and hands-on problem solving to make abstract STEM concepts more accessible to young learners.
- Designed learning experiences that encouraged students to explore, represent, and solve complex problems through multiple forms of reasoning.

Licensure: Elementary School Teacher License, Grade II (2019) and Grade I (2023), Ministry of Education, South Korea

SERVICE

Service to the Profession

2026–Present Graduate Student Representative, SIG-Instructional Design, SITE

2025 Conference Volunteer, AECT International Convention

2023 Expert Reviewer, Elementary Mathematics Workbooks, Educational Broadcasting System (EBS)

Conference Proposal Reviews

2025 Association for Educational Communications and Technology (AECT)

2025 Society for Information Technology and Teacher Education (SITE)

2024 Instructional Systems Technology (IST) Conference

Service to the Program

2026–2027 President, Graduate Students in Instructional Systems Technology (GIST), Indiana University

2025–2026 Vice President of Conference Planning and Conference Chair, GIST, Indiana University

2024–2025 Vice President of Communication, GIST, Indiana University

2021–2023 Graduate Student Representative, Seoul National University of Education

Service to the Community

2024 Fall-2026 Fall Volunteer, STEAM night for the Monroe County Community School Corporation, The Academy of Science and Engineering (ASE), Bloomington, IN

2025 Fall, 2026 Fall CSforGood Project-Based Learning Facilitator, Lakeview, Childchool, Unionville Elementary School, Bloomington, IN

FELLOWSHIPS AND AWARDS

2026–2027 Mary Margaret and Denzil Webb IST Fellowship, Indiana University
2026–2027 Jerrold E. Kemp IST Fellowship, Indiana University
2025–2026 Jerrold E. Kemp IST Fellowship, Indiana University
2025–2026 L.C. Larson Fund, Indiana University
2025–2026 Mary Margaret and Denzil Webb IST Fellowship, Indiana University
2024–2025 Proffitt/DeVault Fellowship, Indiana University
2023 Outstanding STEAM Researcher Award, KOFAC
2022 Top 10 Team Nationwide, ITDA Online Materials Project, KERIS
2022 3rd Prize, Poster Presentation, SNUE Academic Festival
2020 Best E-Class of the Year, KERIS